

10. Given a file of N employee records with a set k of keys (4-digit) which uniquely determine the records in file F . Assume that file F is maintained in memory by a Hash Table (HT) of m memory locations with L as the set of memory address (2-digit) of locations in HT. Let the keys in k and addresses in L are integers.

Design and develop a program in C that uses Hash function $H: k \rightarrow L$ as $H(k) = k \bmod m$ (remainder method) and implement hashing technique to map a given key k to the address space L . Resolve the collision (if any) using Linear probing.

Pseudocode:

Read size of hash table as m

Read number of keys as n

Set

hashTable[i] = -1 For all $i = 0$ to $m-1$

for each key
 $index = key \bmod m$
 if hashtable[index] is empty set hashtable[index] = key
 else $i = 1$
 while hashtable[(index + i) mod m] is not empty
 $i = i + 1$
 hashtable[(index + i) mod m] = key
 for $i = 0$ to $m - 1$
 if hashtable[i] is empty print empty
 else print hashtable[i]

code:

```
#include <stdio.h>
#define MAX 20
int hashtable[MAX];
int m;

void insert(int key) {
    int index = key % m;
    if (hashtable[index] == -1) {
        hashtable[index] = key;
    } else {
        int i = 1;
        while (hashtable[(index + i) % m] != -1) {
            i++;
        }
        hashtable[(index + i) % m] = key;
    }
}

void display() {
    printf("\n Hash Table: \n");
    for (int i = 0; i < m; i++) {
        if (hashtable[i] != -1)
            printf("Address %d : %d \n", i, hashtable[i]);
        else
            printf("Address %d : Empty \n", i);
    }
}

int main() {
    int n, key;
    printf("Enter size of hash table (m): ");
    scanf("%d", &m);
    printf("Enter number of employee records: ");
```



```

scanf("%d", &n);
for (int i=0; i<n; i++)
    hashtable[i] = -1;
printf("Enter %d employee keys (4-digits): (n)", n);
for (int i=0; i<n; i++) {
    scanf("%d", &key);
    insert(key);
}
display();
return 0;
}

```

output:

Enter size of hash table (m) : 10
 Enter number of employee records : 6
 Enter 6 employee keys (4-digits):
 1230
 1247
 1357
 1789
 1999
 1555

Hash table:

Address 0 : 1230
 Address 1 : 1999
 Address 2 : Empty
 Address 3 : Empty
 Address 4 : Empty
 Address 5 : 1555
 Address 6 : Empty
 Address 7 : 1247
 Address 8 : 1357
 Address 9 : 1789